Project Update Report

Course Code: CSE 115 (Section: 4)
Faculty: MSRB
Group Project Number: 7
Project Topic: Personal Diary Management System

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Abstract

This report presents an overview of the current design and development progress of our Personal Diary Management System program in the C programming language. The system provides a simple, console-based platform for users to add, edit, delete and search entries. It also utilizes file handling for data storage and encryption for privacy. This project follows a modular programming approach by dividing different operations into separate functions. This update describes the features, programming techniques and current progress of the project.

1. Introduction

A personal diary is a useful way of recording daily thoughts and activities. Our project aims to create a Personal Diary Management System with a menu-driven console interface. Each diary entry contains a date, title, and content, which can be stored and accessed whenever required.

We are making use of several programming concepts such as functions, structures, file handling and so on. Our code is also using header files such as <stdio.h>, <stdlib.h>, <string.h>, <time.h> and <ctype.h>. The functions we are defining are called in main() to make the program easy to write, understand and debug.

2. Methodology

The project follows a modular programming approach in which the overall diary system is divided into smaller and independent functions.
• **Data Structure:** A custom struct Entry is used to organize the information of each diary entry. It stores the date, title, and content of an entry. This structure allows related data to be handled together and makes file storage easier.

• **Modular Functions:** Separate functions are used for different operations, including add_entry(), view_entries(), search_entries(), edit_entry(), delete_entry(), login(), and change_password(). Helper functions such as read_line(), read_int(), and pause_screen() are also used to keep the main program organized.

• **File Handling**: Diary entries are stored permanently in a binary file named diary.dat. The program uses functions such as fopen(), fread(), fwrite(), and fclose() to read and write diary data. A temporary file is used during editing and deleting operations to safely recreate the updated diary file.

• **Password and Data Protection:** The system includes a login mechanism with a maximum of three password attempts. Passwords and diary titles and contents are processed using a basic XOR encryption technique with a predefined key before being stored in files.

The overall workflow of the system can be represented as: **Login → Main Menu → Select Operation → Process Entry → Save/Display Data → Return to Main Menu**

All our code and related files are stored and shared on GitHub.

3. Task Division

Helper Functions - Jannati Anisha Bonna
Login - Zara Mamnoon
Diary Features - Sanjida Rahman Barsha & Jinnat Jahan Juthi
Main - Elma Shahrin Khan

4. Current Progress

DONE - Identify the structures, variables and constants to be used

DONE - Identify the header files to be used

DONE - Map out the design of the program

DONE - Divide the operations into functions

IN PROGRESS - Writing the code for the functions

IN PROGRESS - Call the functions in main()

TO BE DONE - Debugging and testing

TO BE DONE - Finish the final product

5. Conclusion

We are currently in the design and planning phase of the project. While we have started writing code for the modules, our main focus remains on designing the simplest and most efficient program for the Diary Management System.

We continue to experiment with different programming techniques and understand how to operate in GitHub, all while maintaining proper communication between all the group members.

6. References

Help has been taken from Artificial Intelligence tools such as ChatGPT and Claude in the implementation for code, but the code has been written with changes made manually by each member of the group.